

Synthetic Thermal Load Generator

A tool for generating design-oriented, location-specific synthetic thermal load profiles based on air shade temperature and solar radiation, for use in finite element analysis of structures under environmental conditions.

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Version 1.1.0

<https://doi.org/10.5281/zenodo.20012472>

May 2026

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Purpose and Scope

The Synthetic Thermal Load Generator is a computational tool developed to generate design-oriented, location-specific synthetic thermal load profiles for structural applications. The tool combines air shade temperature and solar radiation to produce representative thermal actions under environmental conditions, enabling the definition of thermal loading scenarios without the need for extensive site-specific monitoring data. The generated thermal load profiles are intended for use in finite element analyses, supporting the evaluation of temperature effects in structures under serviceability conditions. The tool allows users to define environmental parameters and generate consistent thermal actions that can be directly integrated into numerical models. The methodology implemented in the tool is based on the definition of representative thermal cycles, including heating and cooling phases, designed to capture envelope thermal conditions relevant for engineering applications, as described in the supporting scientific publication available at:

<https://doi.org/10.1016/j.tws.2026.115075>

Key Features

- Generation of synthetic air shade temperature profiles
- Generation of synthetic solar radiation profiles
- Definition of heating and cooling phases
- Location-specific input parameters
- Export of structured data for numerical modelling

Intended Use

- Preliminary design assessments
- Parametric thermal studies
- Input generation for finite element models
- Evaluation of environmental thermal actions in composite structures

Important Note

The generated thermal loads are synthetic and design-oriented, aiming to represent envelope conditions rather than exact site-specific measurements. Users are responsible for ensuring that the selected input parameters are consistent with the intended application.

Interface Overview

The graphical user interface of the Synthetic Thermal Load Generator is organised into two main sections corresponding to the thermal load components: solar radiation and air shade temperature. Each component can be activated independently, allowing flexible definition of the generated thermal loads according to the intended application. An overview of the interface is presented in Figure 1.

The screenshot shows the 'Synthetic Thermal Load Generator' window. It features two main sections: 'Solar Radiation' and 'Air Shade Temperature'. The 'Solar Radiation' section is currently selected, indicated by a red box labeled '1' next to its radio button. Below it, the 'Solar Radiation Settings' section is expanded, with a red box labeled '2' next to its title. This section contains input fields for Latitude, Longitude, Time Zone (set to UTC±0), Solar Constant (1366.1), Clearness Index (Cooling: 0.2, Heating: 0.8), and Interval (seconds) for Radiation. There are also checkboxes for 'Inclined Surfaces', 'Reflected Radiation', and 'Generate Solar Position Inputs for Self-Shadowing Analysis'. The 'Air Shade Temperature' section is below, with a red box labeled '3' next to its radio button. Its settings section, labeled '4', includes input fields for Minimum Temperature, Maximum Temperature, Daily Temperature Variation, and Interval (seconds) for Temperature. A red box labeled '5' points to the 'Generate' button. At the bottom, there are logos for 'ISISE' (Institute for Sustainability and Innovation in Structural Engineering), 'fct' (Fundação para a Ciência e a Tecnologia), and 'University of Minho School of Engineering'. Two buttons at the bottom are labeled 'View Supporting Publication' (with a red box labeled '6') and 'Contact & Feedback' (with a red box labeled '7').

Synthetic Thermal Load Generator

☐ Solar Radiation

Solar Radiation Settings

Latitude: ⓘ

Longitude: ⓘ

Time Zone: UTC±0 (London, Lisbon, Greenwich) ⓘ

Solar Constant (W/m²): 1366.1 ⓘ

Clearness Index (Cooling): 0.2 ⓘ

Clearness Index (Heating): 0.8 ⓘ

Interval (seconds) for Radiation: ⓘ

☐ Inclined Surfaces ⓘ

☐ Reflected Radiation ⓘ

☐ Generate Solar Position Inputs for Self-Shadowing Analysis ⓘ

☐ Air Shade Temperature

Air Shade Temperature Settings

Minimum Temperature (°C): ⓘ

Maximum Temperature (°C): ⓘ

Daily Temperature Variation (°C): ⓘ

Interval (seconds) for Temperature: ⓘ

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Figure 1. Interface overview.

The main elements of the interface are identified as follows:

- (1) Solar Radiation selection checkbox, used to activate or deactivate the solar radiation component.
- (2) Solar Radiation Settings panel, where the parameters required for the generation of synthetic solar radiation are defined.
- (3) Air Shade Temperature selection checkbox, used to activate or deactivate the air temperature component.
- (4) Air Shade Temperature Settings panel, where the parameters defining the synthetic air temperature cycles are introduced.
- (5) Generate button, which executes the generation of the synthetic thermal load profiles based on the defined inputs.
- (6) Supporting publication button, providing access to the scientific publication underlying the methodology implemented in the tool.
- (7) Contact & Feedback button, enabling users to report issues or provide feedback.

Solar Radiation Workflow

The generation of synthetic solar radiation profiles follows a structured workflow involving the definition of environmental parameters, optional surface configuration, and execution of the generation process. The main steps are outlined below.

1. Activate the solar radiation component.
2. Define environmental and radiation parameters.
3. Configure inclined surfaces (optional).
4. Define reflected radiation parameters (optional).
5. Choose whether to generate the Self-Shadowing Analysis input files.
6. Run the generation process.

STEP 1 — Activating Solar Radiation

The solar radiation component is activated by selecting the **Solar Radiation** checkbox. Once activated, the corresponding input fields become available.

The screenshot shows the 'Synthetic Thermal Load Generator' application window. The 'Solar Radiation' checkbox is checked, and its settings are visible. Below it, the 'Air Shade Temperature' checkbox is unchecked. The interface includes a 'Generate' button and logos for ISISE, fct, and the University of Minho. At the bottom, there are links for 'View Supporting Publication' and 'Contact & Feedback'.

Synthetic Thermal Load Generator

☒ **Solar Radiation**

Solar Radiation Settings

Latitude: ⓘ

Longitude: ⓘ

Time Zone: UTC±0 (London, Lisbon, Greenwich) ▼ ⓘ

Solar Constant (W/m²): ⓘ

Clearness Index (Cooling): ⓘ

Clearness Index (Heating): ⓘ

Interval (seconds) for Radiation: ⓘ

☐ Inclined Surfaces ⓘ

☐ Reflected Radiation ⓘ

☐ Generate Solar Position Inputs for Self-Shadowing Analysis ⓘ

☐ **Air Shade Temperature**

Air Shade Temperature Settings

Minimum Temperature (°C): ⓘ

Maximum Temperature (°C): ⓘ

Daily Temperature Variation (°C): ⓘ

Interval (seconds) for Temperature: ⓘ

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Figure 2. Main interface with solar radiation component activated.

STEP 2 — Defining Input Parameters

- The required environmental parameters must be defined, including:
- Latitude and longitude (in decimal degrees)
- Time zone
- Solar constant
- Clearness index values for heating and cooling phases (a value between 0 and 1, recommended values are 0.2 for cooling and 0.8 for heating)
- Time interval for radiation calculations (in seconds)

STEP 3 — Defining Inclined Surfaces (Optional)

Inclined surfaces can be defined to evaluate solar radiation on non-horizontal elements.

By selecting the **Inclined Surfaces** option, a dedicated window is opened (Figure 3a), where surfaces can be added, edited, or removed.

Each surface is defined by (Figure 3b):

- A user-defined name
- Azimuth angle
- Inclination angle

The defined surfaces are stored and listed (Figure 3c), allowing multiple configurations to be managed.

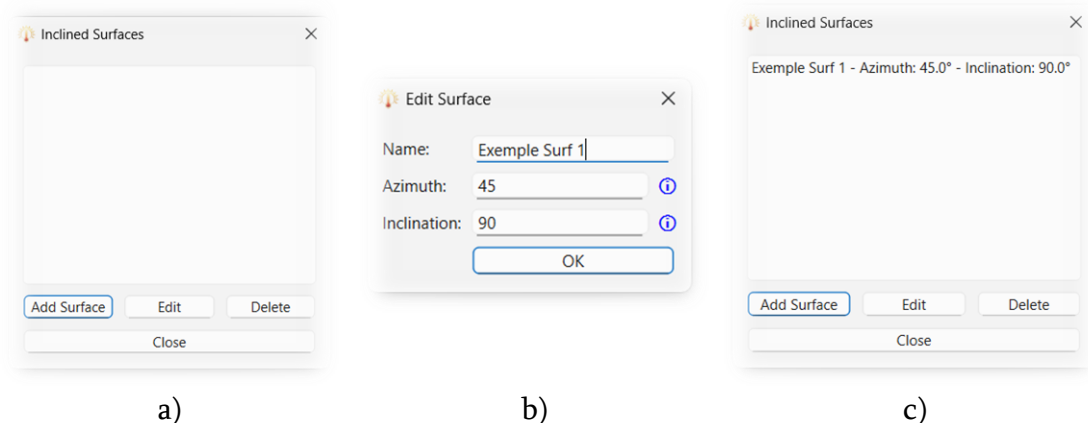


Figure 3. Inclined surfaces configuration window: a) overview; b) surface parameters; c) added surfaces overview.

As illustrated in Figure 4, the inclination angle is measured as the angle between the surface and the horizontal plane, following the outward-facing direction of the surface.

This definition implies that the inclination is inherently opposite to the direction of the incoming solar rays.

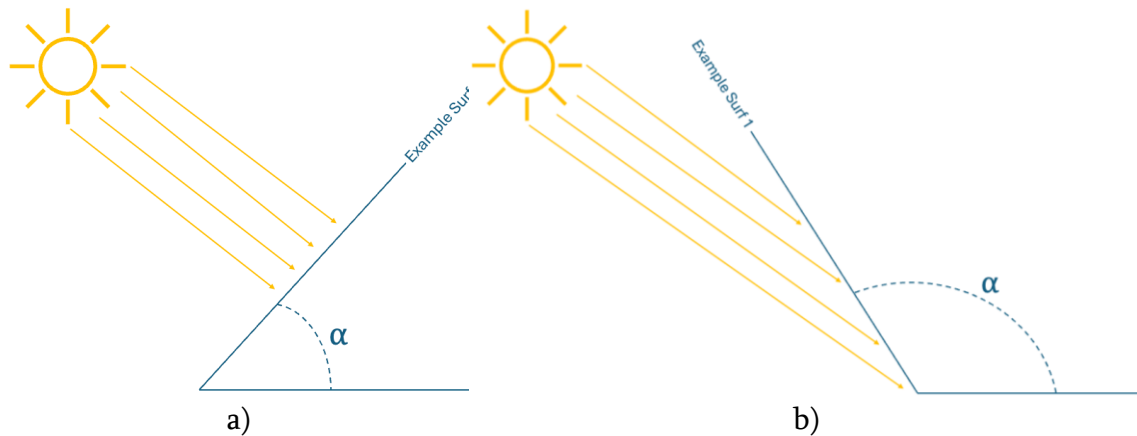


Figure 4. Definition of the inclination angle relative to the surface orientation and incoming solar radiation: a) acute angle; b) obtuse angle.

STEP 4 — Defining Reflected Radiation (Optional)

Reflected radiation effects can be included on individual surfaces.

Through the **Reflected Radiation** option (Figure 5a), users can specify:

- The target surface (Figure 5b)
- Albedo (Figure 5c)
- Inclination (Figure 5c)
- Source of reflection (e.g., ground or another surface) (Figure 5c)

The defined reflected configurations are stored and listed (Figure 5d), allowing multiple configurations to be managed.

Note: Bottom Surface as surface and Ground as source are standard options always available without the need to be created.

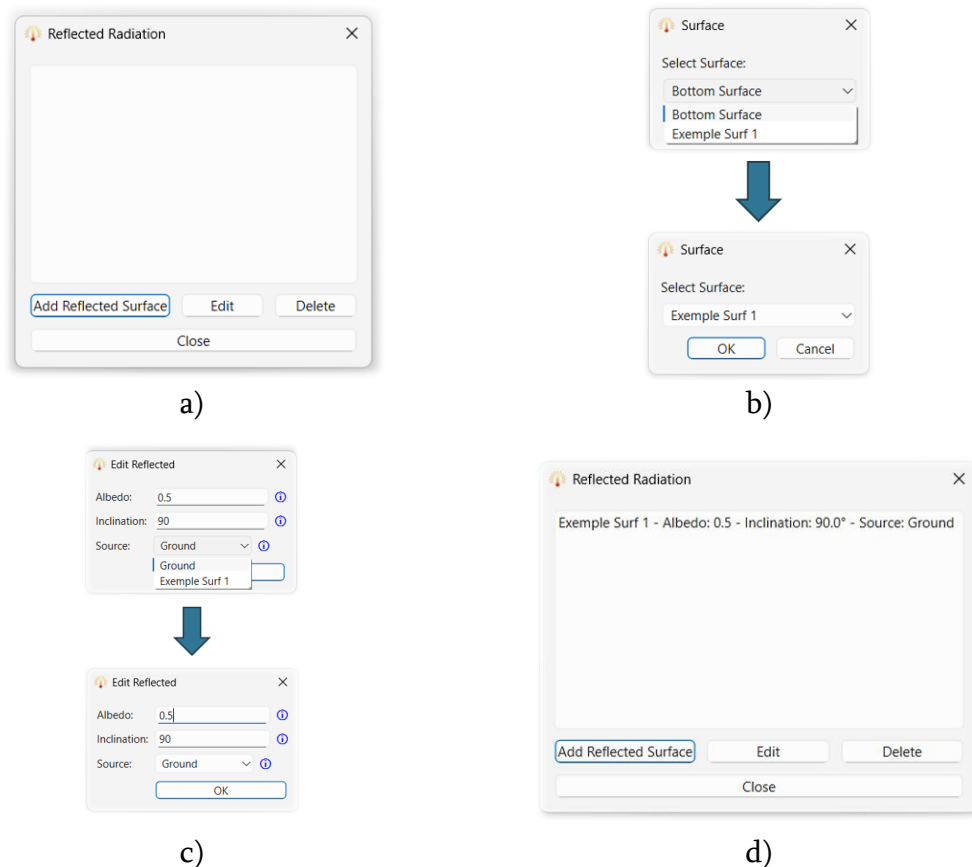


Figure 5. Reflected radiation configuration: a) overview; b) surface selection; c) definition of reflected radiation parameters; d) added reflected radiation configuration.

STEP 5 — Self-Shadowing Input (Optional)

The option **Generate Solar Position Inputs for Self-Shadowing Analysis** enables the export of solar position data (time, hour angle, declination) for use in self-shadowing analyses, shading models, or advanced post-processing (using the Self-Shadowing Analysis tool <https://doi.org/10.5281/zenodo.20012999>) or another compatible external tool).

STEP 6 — Running the Generation

After defining all required parameters, the synthetic solar radiation profiles are generated by clicking the **Generate** button. After completion, the generated CSV files are saved in the selected output directory.

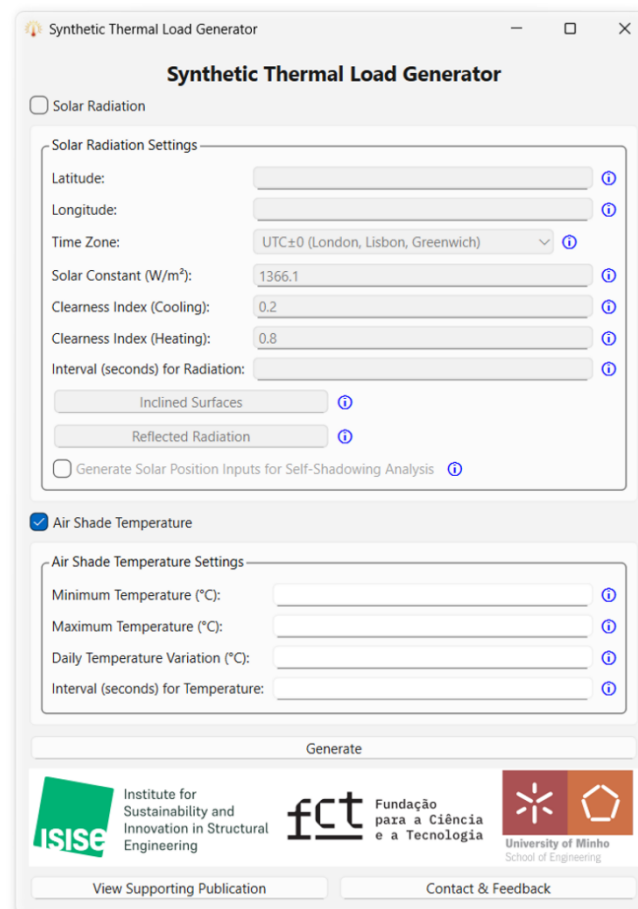
Air Shade Temperature Workflow

The generation of synthetic air shade temperature profiles is based on the definition of temperature limits and daily variations, allowing the construction of representative thermal cycles. The workflow is simpler than that of solar radiation and focuses on the definition of temperature bounds and temporal resolution.

1. Activate the air shade temperature component
2. Define temperature parameters.
3. Run the generation process.

STEP 1 — Activating Air Shade Temperature

The air shade temperature component is activated by selecting the **Air Shade Temperature** checkbox. Once activated, the corresponding input fields become available.



The image shows a software window titled "Synthetic Thermal Load Generator". It has two main sections: "Solar Radiation" and "Air Shade Temperature".

The "Solar Radiation" section is currently inactive, indicated by an unchecked checkbox. It contains the following settings:

- Latitude: [text input]
- Longitude: [text input]
- Time Zone: [dropdown menu showing "UTC±0 (London, Lisbon, Greenwich)"]
- Solar Constant (W/m²): [text input with value "1366.1"]
- Clearness Index (Cooling): [text input with value "0.2"]
- Clearness Index (Heating): [text input with value "0.8"]
- Interval (seconds) for Radiation: [text input]
- Inclined Surfaces: [checkbox]
- Reflected Radiation: [checkbox]
- Generate Solar Position Inputs for Self-Shadowing Analysis: [checkbox]

The "Air Shade Temperature" section is active, indicated by a checked checkbox. It contains the following settings:

- Minimum Temperature (°C): [text input]
- Maximum Temperature (°C): [text input]
- Daily Temperature Variation (°C): [text input]
- Interval (seconds) for Temperature: [text input]

At the bottom of the window, there is a "Generate" button. Below the button are logos for "ISISE" (Institute for Sustainability and Innovation in Structural Engineering), "fct" (Fundação para a Ciência e a Tecnologia), and "University of Minho School of Engineering". At the very bottom, there are two buttons: "View Supporting Publication" and "Contact & Feedback".

Figure 6. Main interface with temperature component activated.

STEP 2 — Defining Temperature Parameters

- The **minimum and maximum air shade temperatures** must be defined. These values represent the lower and upper bounds of the synthetic thermal cycle and should be selected according to the environmental conditions of the location under consideration.
- The **daily temperature variation** defines the amplitude of the temperature fluctuation within each cycle. This parameter controls the difference between daily minimum and maximum values and should be consistent with design assumptions or relevant standards.
- The **time interval** determines the temporal resolution of the generated temperature profile. Smaller intervals provide more detailed results but may increase computational cost and output size.

STEP 3 — Running the Generation

After defining all required parameters, the synthetic air shade temperature profiles are generated by clicking the **Generate** button. After completion, the generated CSV files are saved in the selected output directory.

Output Files and Structure

The Synthetic Thermal Load Generator produces structured output files containing the generated thermal load data. The outputs are organised into separate files corresponding to radiation and temperature components, as well as optional self-shadowing inputs.

The generated outputs are organised into multiple files (Figure 7), including:

- Radiation_Heating.csv
- Radiation_Cooling.csv
- Temperature_Heating.csv
- Temperature_Cooling.csv
- Self_Shadowing_Inputs_Heating.csv (*optional*)
- Self_Shadowing_Inputs_Cooling.csv (*optional*)

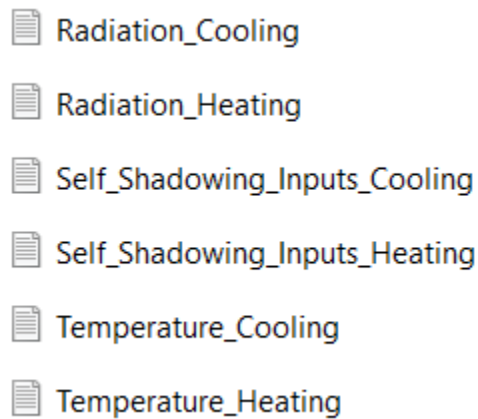


Figure 7. Example of generated output files.

Note: The generation of self-shadowing input files depends on the activation of the corresponding option in the interface.

Radiation Output Structure

The radiation output files contain time-dependent values of solar radiation components (Figure 8).

The following columns are always generated:

- Time (s)
- Global Radiation (W/m^2)
- Diffuse Radiation (W/m^2)

Additional columns are generated depending on the user-defined configuration:

- One column for each defined inclined surface (direct radiation on the surface)
- One column for each reflected radiation component

Time (s)	Global Radiation (W/m^2)	Diffuse Radiation (W/m^2)	Inclined Radiation Exemple Surf 1 (W/m^2)	Reflected Radiation Exemple Surf 1 from Ground (W/m^2)
0	0	0	0	0
30	0	0	0	0
60	0	0	0	0
90	0	0	0	0
120	0	0	0	0
150	0	0	0	0
180	0	0	0	0

Figure 8. Structure of the generated solar radiation output.

Note: The surface naming of additional columns reflects the corresponding names defined by the user.

Temperature Output Structure

The temperature output files contain the synthetic air shade temperature profiles as a function of time (Figure 9).

The output is organised into two columns:

- Time (s)
- Temperature (°C)

Time (s)	Temperature (°C)
0	-5
30	-5
60	-4.9999
90	-4.9999
120	-4.9998
150	-4.9996
180	-4.9995
210	-4.9993

Figure 9. Structure of the generated air shade temperature output.

Note: The generated temperature profiles reflect the defined minimum and maximum limits, as well as the specified daily temperature variation.

Self-Shadowing Input Files

When the self-shadowing option is activated, additional files containing solar position data are generated.

These files include:

- Time (s)
- Hour angle (°)
- Declination (°)

The generated data can be used as input for external self-shadowing analyses (using the Self-Shadowing Analysis tool <https://doi.org/10.5281/zenodo.20012999>) or custom radiation models.

Time (s)	Hour Angle (°)	Declination (°)
0	169.3831156	-20.8471721
30	169.5081156	-20.8471721
60	169.6331156	-20.8471721
90	169.7581156	-20.8471721
120	169.8831156	-20.8471721
150	170.0081156	-20.8471721
180	170.1331156	-20.8471721
210	170.2581156	-20.8471721

Figure 10. Structure of the generated self-shadowing input file output.

Filled-out example

Figure 11 presents a filled-out example including both solar radiation and air shade temperature components. The selected parameters correspond to a representative case where radiation, temperature, and optional self-shadowing inputs are generated simultaneously.

The image shows a software window titled "Synthetic Thermal Load Generator". It contains two main sections, each with a checked checkbox and a set of input fields.

Solar Radiation (checked):

- Solar Radiation Settings:**
 - Latitude: 41.45
 - Longitude: -8.29
 - Time Zone: UTC±0 (London, Lisbon, Greenwich)
 - Solar Constant (W/m²): 1366.1
 - Clearness Index (Cooling): 0.2
 - Clearness Index (Heating): 0.8
 - Interval (seconds) for Radiation: 30
 - ☐ Inclined Surfaces
 - ☐ Reflected Radiation
- ☒ Generate Solar Position Inputs for Self-Shadowing Analysis

Air Shade Temperature (checked):

- Air Shade Temperature Settings:**
 - Minimum Temperature (°C): -5
 - Maximum Temperature (°C): 40
 - Daily Temperature Variation (°C): 10
 - Interval (seconds) for Temperature: 30

At the bottom of the settings area is a "Generate" button. Below this is a footer section with logos for ISISE (Institute for Sustainability and Innovation in Structural Engineering), fct (Fundação para a Ciência e a Tecnologia), and the University of Minho School of Engineering. At the very bottom are two buttons: "View Supporting Publication" and "Contact & Feedback".

Figure 11. Filled-out example